Multivariate Calculus Lecture Notes
Section 13.2 – Calculus of Vector-Valued Functions

I. Limits and vector functions
   a. If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, then

$$\lim_{t\to a} \mathbf{r}(t) = \lim_{t\to a} f(t)\mathbf{i} + \lim_{t\to a} g(t)\mathbf{j} + \lim_{t\to a} h(t)\mathbf{k},$$

   provided these limits exist.
   b. Example: $\lim_{t\to 0} \mathbf{r}(t)$, where $\mathbf{r}(t) = (1+t^3)\mathbf{i} + te^{-t}\mathbf{j} + \dfrac{\sin t}{t}\mathbf{k}$

$$\lim_{t\to 0} \mathbf{r}(t) = \lim_{t\to 0}(1+t^3)\mathbf{i} + \lim_{t\to 0} te^{-t}\mathbf{j} + \lim_{t\to 0} \frac{\sin t}{t}\mathbf{k} = \mathbf{i} + \mathbf{k}$$

II. Continuity of vector functions
   a. A vector-valued function $\mathbf{r}(t) = f(t)\mathbf{i}+g(t)\mathbf{j}+h(t)\mathbf{k}$ is *continuous* at t_0 if $\lim_{t\to t_0} \mathbf{r}(t) = \mathbf{r}(t_0)$.
   b. in other words, $\mathbf{r}(t)$ is continuous at t_0 if and only if $f(t), g(t)$, and $h(t)$ are continuous at t_0.

III. Differentiation of vector functions
   a. Definition
      i. The *derivative* of a vector-valued function $\mathbf{r}(t)$ is defined by

$$\mathbf{r}'(t) = \lim_{h\to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}.$$

      ii. This is differentiation with respect to t, not x, y, or z.
      iii. If $\mathbf{r}'(t)$ exists, then $\mathbf{r}$ is said to be *differentiable* at t.
   b. Differentiation rules
      i. The derivative of $\mathbf{r}(t)$ follows all the usual differentiation rules.
      ii. In particular, $\dfrac{d}{dt}\mathbf{r}(t) = \dfrac{d}{dt}(f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}$.
      iii. How does the derivative behave in the presence of the dot and cross products?
         1. $\dfrac{d}{dt}\left(\mathbf{u}(t)\cdot\mathbf{v}(t)\right) = \mathbf{u}'(t)\cdot\mathbf{v}(t) + \mathbf{u}(t)\cdot\mathbf{v}'(t)$.
         2. $\dfrac{d}{dt}\left(\mathbf{u}(t)\times\mathbf{v}(t)\right) = \mathbf{u}'(t)\times\mathbf{v}(t) + \mathbf{u}(t)\times\mathbf{v}'(t)$.
         3. See text for proofs.
   c. Geometric interpretation of $\mathbf{r}'(t)$
      i. Graphically, here's what's going on with $\mathbf{r}(t+\Delta t) - \mathbf{r}(t)$:

      ii. Dividing by Δt merely changes the length of the vector.
      iii. As $\Delta t \to 0$, $\mathbf{r}'(t)$ approaches a vector tangent to the direction of motion:

iv. Even if Δt is negative, this is what happens.

d. Examples
   i. Graph $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j}$ in the plane, find its derivative, and show the value of the derivative on the graph for a couple of values.
      1. Points
         - $\mathbf{r}(0) = \mathbf{i}$
         - $\mathbf{r}(\pi/2) = \mathbf{j}$
         - $\mathbf{r}(\pi) = -\mathbf{i}$
         - $\mathbf{r}(3\pi/2) = -\mathbf{j}$
      2. Graphically:

      3. $\mathbf{r}'(t) = -\sin t\mathbf{i} + \cos t\mathbf{j}$.
      4. Derivative points
         - $\mathbf{r}'(0) = \mathbf{j}$
         - $\mathbf{r}'(\pi/2) = -\mathbf{i}$
      5. Derivative, graphically:

   ii. Find an equation of the line tangent to the graph of $\mathbf{r}(t) = (6 - t^2)\mathbf{i} + e^{3t}\mathbf{j} - \arctan t\mathbf{k}$ when $t = 0$.
      1. We need a point and a direction vector.
      2. Point: $\mathbf{r}(0) = (6, 1, 0)$.
      3. Direction vector
         - $\mathbf{r}'(t) = -2t\mathbf{i} + 3e^{3t}\mathbf{j} - \dfrac{1}{1+t^2}\mathbf{k}$
         - $\mathbf{r}'(0) = 3\mathbf{j} - \mathbf{k}$
      4. Thus an equation of the line is $\mathbf{L}(t) = \langle 6, 1 + 3t, -t\rangle$.

e. What happens when $\mathbf{r}'(t) = \mathbf{0}$
   i. What is $\mathbf{r}'(t)$ at a local max or min?
      1. Graphically:

      2. $\mathbf{r}'(t)$ is a vector in the x direction.
      3. In other words, the $\mathbf{j}$ component of $\mathbf{r}'(t)$ is zero.
   ii. Similarly, if the $\mathbf{i}$ component of $\mathbf{r}'(t)$ is zero then the graph of $\mathbf{r}(t)$ has a vertical tangent.
   iii. So what does it mean when $\mathbf{r}'(t) = \mathbf{0}$?
      1. It means that the curve has no (well-defined) tangent at that point.
      2. For example, consider $\mathbf{r}(t) = (1 + t^3)\mathbf{i} + t^2\mathbf{j}$ in the plane.
         - Points
           - $\mathbf{r}(0) = \langle 1, 0\rangle$
           - $\mathbf{r}(1) = \langle 2, 1\rangle$
           - $\mathbf{r}(-1) = \langle 0, 1\rangle$

  - $\mathbf{r}(2) = \langle 9, 4\rangle$
  - $\mathbf{r}(-2) = \langle -7, 4\rangle$
- Graphically:

- $\mathbf{r}'(t) = 3t^2\mathbf{i} + 2t\mathbf{j}$
- $\mathbf{r}'(t) = \mathbf{0}$ when $t = 0$.

3. Generally, but not always, when $\mathbf{r}'(t) = \mathbf{0}$ the graph of $\mathbf{r}(t)$ has a cusp.

iv. A vector-valued function $\mathbf{r}$ is said to be *smooth* on an open interval I if $\mathbf{r}'$ is continuous on I and $\mathbf{r}'(t) \neq \mathbf{0}$ for any value of t in I.

IV. Integration

a. Integration, like differentiation, is done component-wise.

b. Definite integration: If f, g, and h are integrable over $[a, b]$, then

$$\int_a^b \mathbf{r}(t)\,dt = \left(\int_a^b f(t)\,dt\right)\mathbf{i} + \left(\int_a^b g(t)\,dt\right)\mathbf{j} + \left(\int_a^b h(t)\,dt\right)\mathbf{k}.$$

c. Indefinite integration

i. Suppose $\mathbf{R}(t)$ is some vector-valued function such that $\mathbf{R}'(t) = \mathbf{r}(t)$.

ii. Then $\mathbf{R}(t) = F(t)\mathbf{i} + G(t)\mathbf{j} + H(t)\mathbf{k}$ for some functions F, G, and H with the property that $F'(t) = f(t)$, $G'(t) = g(t)$, and $H'(t) = h(t)$.

iii. We have

$$\begin{aligned}\int \mathbf{r}(t)\,dt &= \left(\int f(t)\,dt\right)\mathbf{i} + \left(\int g(t)\,dt\right)\mathbf{j} + \left(\int h(t)\,dt\right)\mathbf{k}\\ &= (F(t) + C_1)\,\mathbf{i} + (G(t) + C_2)\,\mathbf{j} + (H(t) + C_3)\,\mathbf{k}\\ &= [F(t)\mathbf{i} + G(t)\mathbf{j} + H(t)\mathbf{k}] + [C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}]\\ &= \mathbf{R}(t) + \mathbf{C}.\end{aligned}$$

iv. So the constant of integration is a vector $\mathbf{C}$.

d. Examples

i. $\mathbf{r}(t) = 2\cos t\mathbf{i} + \sin t\mathbf{j} + 2t\mathbf{k}$ integrated from 0 to $\pi/2$.

$$\begin{aligned}\int_0^{\pi/2} \mathbf{r}(t)\,dt &= \left(\int_0^{\pi/2} 2\cos t\,dt\right)\mathbf{i} + \left(\int_0^{\pi/2} \sin t\,dt\right)\mathbf{j} + \left(\int_0^{\pi/2} 2t\,dt\right)\mathbf{k}\\ &= \left[2\sin t\mathbf{i} - \cos t\mathbf{j} + t^2\mathbf{k}\right]_0^{\pi/2} = (2-0)\mathbf{i} - (0-1)\mathbf{j} + (\pi^2/4 - 0)\mathbf{k}\\ &= 2\mathbf{i} + \mathbf{j} + \pi^2/4\,\mathbf{k}.\end{aligned}$$

ii. Find $\mathbf{r}(t)$, where $\mathbf{r}'(t) = te^{-t^2}\mathbf{i} - e^{-t}\mathbf{j} + \mathbf{k}$, and $\mathbf{r}(0) = \frac{1}{2}\mathbf{i} - \mathbf{j} + \mathbf{k}$. (This is an example of a vector *differential equation.*)

1. $\mathbf{r}(t) = -\frac{1}{2}e^{-t^2}\mathbf{i} + e^{-t}\mathbf{j} + t\mathbf{k} + \mathbf{C}$.
2. Subbing in for the initial condition, we obtain $\frac{1}{2}\mathbf{i} - \mathbf{j} + \mathbf{k} = \mathbf{r}(0) = -\frac{1}{2}\mathbf{i} + \mathbf{j} + \mathbf{C}$.
3. Thus $\mathbf{C} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$.
4. Therefore, $\mathbf{r}(t) = (1 - \frac{1}{2}e^{-t^2})\mathbf{i} + (e^{-t} - 2)\mathbf{j} + (t+1)\mathbf{k}$.

Homework: Section 13.2 #4, 7, 12, 14, 21, 24, 31, 34, 37, 42, 45, 49, 51.